

Def. Let V be a finite-dimensional inner product space over $F = \mathbb{C}$ or $\mathbb{R}$.

A linear operator $T: V \rightarrow V$ satisfying

$$\forall x \in V, \|T(x)\| = \|x\|$$

is called Unitary when $F = \mathbb{C}$, orthogonal when $F = \mathbb{R}$.

Lemma. Let V be an inner product space, $H: V \rightarrow V$ be a Hermitian.

If for all $x \in V$, $\langle x, H(x) \rangle = 0$, then $H = 0$.

Proof. Let $x \in V$ be given. Then,

$$\begin{aligned} 0 &= \langle x + H(x), H(x + H(x)) \rangle = \langle x + H(x), H(x) + H^2(x) \rangle \\ &= \langle x, H(x) \rangle + \langle x, H^2(x) \rangle + \langle H(x), H(x) \rangle + \langle H(x), H^2(x) \rangle \\ &= 0 + \|H(x)\|^2 + 0 \end{aligned}$$

Therefore, $H(x) = 0$.

Thm. Let V be a fin. dim. inn. V.s over $F = \mathbb{C}$ or $\mathbb{R}$.

For any linear operator $T: V \rightarrow V$,

1) For all $x \in V$, $\|T(x)\| = \|x\|$

2) For any $x, y \in V$, $\langle T(x), T(y) \rangle = \langle x, y \rangle$

3) $T \circ T^* = I$

4) $T^* \circ T = I$

Proof. Suppose 1). we follow along $1) \Rightarrow 3) \Rightarrow 4) \Rightarrow 1)$.

Let $x \in V$ be given. Then,

$$\langle x, x \rangle = \langle T(x), T(x) \rangle = \langle x, (T^* \circ T)(x) \rangle \text{ which implies } \langle x, (I - T^* \circ T)(x) \rangle = 0.$$

And, since $(I - T \circ T^*)^* = I^* - (T \circ T^*)^* = I - T^* \circ T$, it is Hermitian, therefore, $I = T^* \circ T$, by the Lemma. Clearly, this implies $I = T \circ T^*$.

Finally, if $T^* \circ T = I$, then for any $x \in V$,

$$\langle x, x \rangle = \langle (T^* \circ T)(x), x \rangle = \langle T(x), T(x) \rangle$$

$$I_n \quad F = \mathcal{C}.$$

